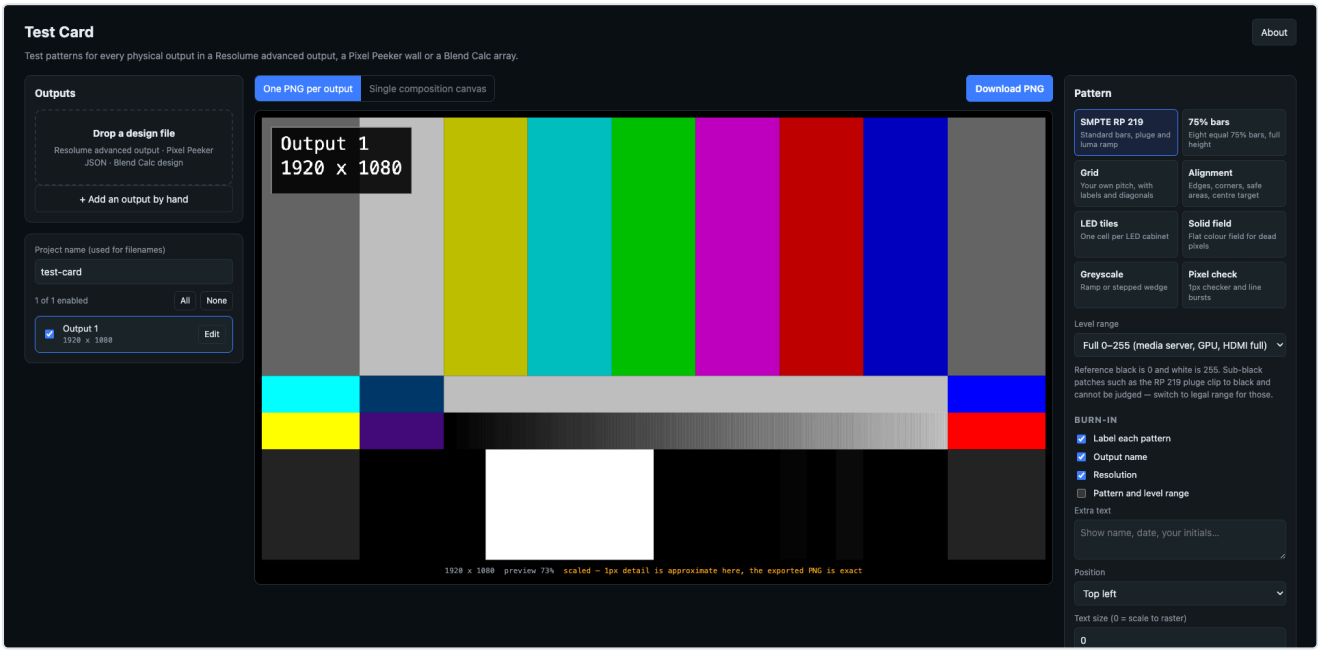
Test Card user guide

Test Card generates test patterns for every physical output in a show, from the file that already describes the outputs.



One 1920 × 1080 output added by hand, SMPTE RP 219 bars at full range with the output name and resolution burnt in. The preview is scaled; the exported PNG is pixel-exact.

Import a **Resolume advanced output**, a **Pixel Peeker** wall or a **Blend Calc** projector array, and get one PNG per physical output at that output's exact raster, or a single composition-sized image that proves the output map is what you think it is. Everything runs in the browser, and nothing is uploaded.

Before you rely on this: the RP 219 pattern is checked against a reference **measured** from a real generator at four resolutions, rather than transcribed from prose descriptions of the standard, and its fifteen colours are computed through Rec. 709 and land exactly on those measured values, which is itself the proof the measurement is sound. The importers are tested against **real files**, not hand-written fixtures, and the whole path has been driven end to end in a browser.

The MP4 loops are made in Chrome and read back by ffmpeg: 75% bars come back within one code value of the standard, and the loops go round with no seam. They've also been played in Resolume Arena 7.28.1, which opens them, gets the colour right and loops them cleanly, with one catch covered under **Moving patterns** below. A loop too big for the browser to encode whole comes out as tiles; decoded and stitched back together, an 8K one matches the picture drawn whole.

The idle patterns are a port of Idler's screensavers, checked mesh for mesh against Idler's own code, and their loops have been encoded and read back with no seam.

Live output has been driven in Chrome 155 with three pretend displays: it lists them, opens a window over each, and each goes full screen on its own display at 1:1. It hasn't been tried with a real second display yet.

No image or loop it produces has been played out to a real projector, LED processor or display and measured. The patterns are correct as pixels; what a screen does with them is something this can't tell you.

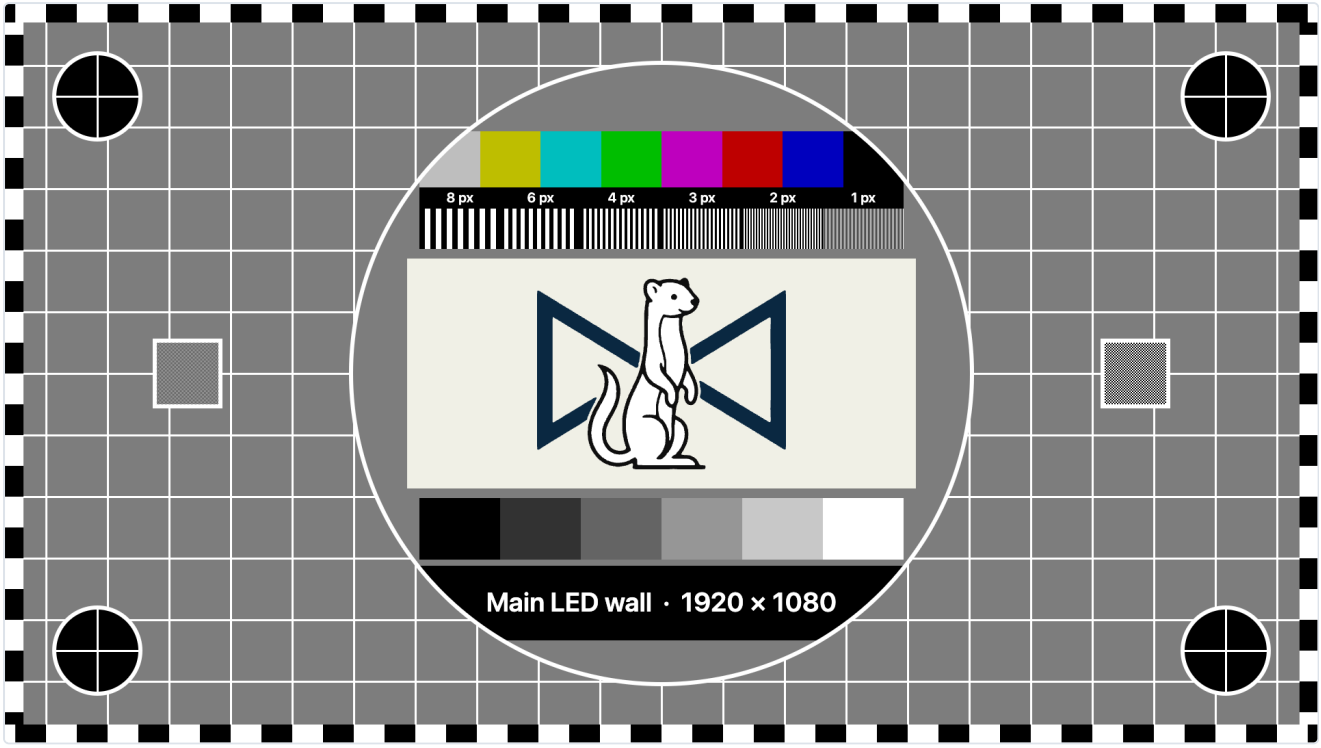
This codebase was created with AI assistance, directed and reviewed by a human author.

The patterns

Pattern	What it is for
SMPTE RP 219	Standard bars, +I/+Q, luma ramp and pluge.
75% bars	Eight equal full-height bars. Survives being squeezed onto a narrow LED strip where RP 219's four rows would be unreadable.
Grid	Your own pitch or division count, heavy lines every N, cell labels, diagonals, centre target.
Alignment	1px edge border, corner markers with pixel counts, safe areas, centre target and circles, 1px checker patches.
LED tiles	One cell per cabinet — either a uniform tile size you state, or the **real cabinet map** imported from Pixel Peeker, including irregular walls, chain order and per-port colouring.
Solid field	Flat colour at a chosen level, for dead pixels and uniformity.
Greyscale	Continuous ramp or stepped wedge, optionally split into R, G, B and neutral.
Pixel check	1px checkerboard and line bursts. **If it renders as flat grey on the real output, something in the chain is scaling.**
Classic test card	A grey grid, castellated edges and a big circle holding colour bars, gratings from 8 px down to 1 px, a picture panel, six greyscale steps and the output's name. See below.
Levels check	Full or limited range, and which end has it wrong. See below.
AV sync	A flash and a beep on the same frame, and a scale that reads the offset in frames. See below.

Every pattern can carry a burn-in (output name, resolution, pattern, plus your own text) and can draw the imported slice or port outlines on top.

The classic test card



Laid out the way the old broadcast cards were, for the same reasons:

- **The circle** and the **corner circles** (one in each corner, where there's room) are round when the aspect ratio is right. An egg means something is stretching the picture.
- **The castellations** run along every edge. Count how many you can see to know how much of the picture overscan or a mask is eating.
- **The gratings** go from 8 px lines down to 1 px. Where they turn to mush is where the resolution runs out; the 1 px one (and the **checker** on the left) goes flat grey through anything that scales.
- **The six steps** run from black to white. Two that look the same at either end mean crushed blacks or clipped whites.
- **The box at the foot** names the output and its raster. Untick **Name the output** and the box goes.

In the middle is the Stoatworks logo by default. Pick **Your image** to use your own picture (the same one the **Your image** pattern uses, kept in this browser and never uploaded), or **Your text** for a line or two of your own: the venue, the screen, the show. The text is sized to fill the panel. Set the panel's **Background**, and the text's colour.

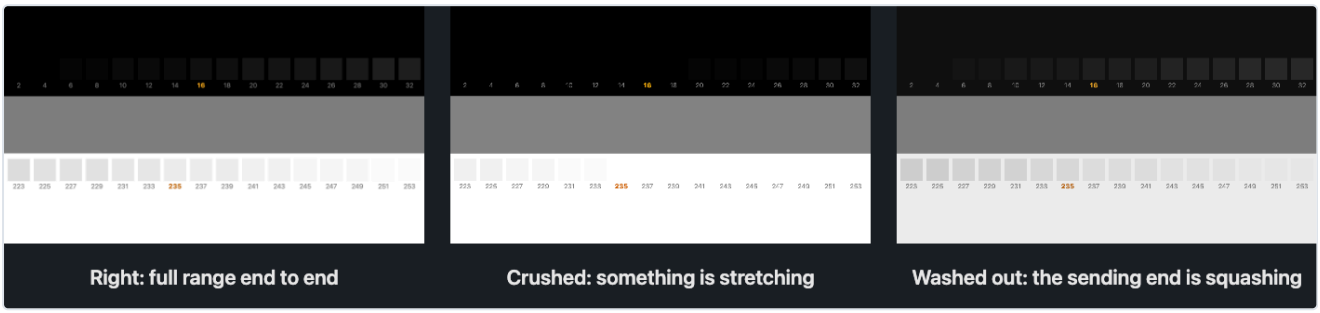
The card is always drawn in full range, whatever the level setting says. For checking levels, use the bars, the greyscale, or the **levels check**.

The levels check

Full or limited range is the setting everybody has an opinion about and nobody can see. A GPU, a media server output, a converter, a processor input and a display can each be set either way, or to

"auto", which means "guess", and the picture only looks right when the two ends of each link agree.

The levels check is a row of near-black squares on black (codes 2 to 32) and a row of near-white squares on white (223 to 253), each labelled, with the limited-range edges (16 and 235) picked out. It's always drawn in full range, since a media server plays a full-range picture.



What you see	What it means	What to change
Every square in both rows, black is black, white is white	Full range end to end (or limited at both ends, which works too)	Nothing
The squares from 16 down and from 235 up have gone, the coloured ones included	**Crushed.** Something is stretching 16–235 out to 0–255, so 16 lands on black, 235 on white and everything beyond clips. Usually the receiving end expecting limited; sometimes a player reading a full-range file as limited, or a converter set to expand	Set the receiver's input to full, or the sender's output to limited
Every square is there, but black is grey and white is dull	**Washed out.** The sending end is squashing to limited, and nothing stretches it back	Set the sender's output to full, or the receiver's input to limited

Crushed shows itself: squares vanish. Washed out doesn't, because nothing is lost, so judge the black against the screen switched off, or against a dark bit of set.

What one picture can't tell you is which box in a longer chain did it. Look at the card on any monitor partway along: a media server's own preview, a processor's monitor out, a scaler's loop-through. The fault is between the last place it looks right and the first place it looks wrong. **Explain the results on the card** prints the table above on the pattern itself, for the person at the other end of the radio. On an output under about 800 pixels wide the notes would be too small to read, so they're left off and only the squares are drawn.

AV sync

Every so often (once a second unless you change **Flash and beep every**) the pattern has a sync frame: the side panels flash white, the hand going round the ring reaches the top, and the sound starts a 1 kHz beep at −18 dBFS that lasts that one frame. Under the ring, boxes numbered −6 to +6 light one per frame either side of it, 0 on the sync frame itself.

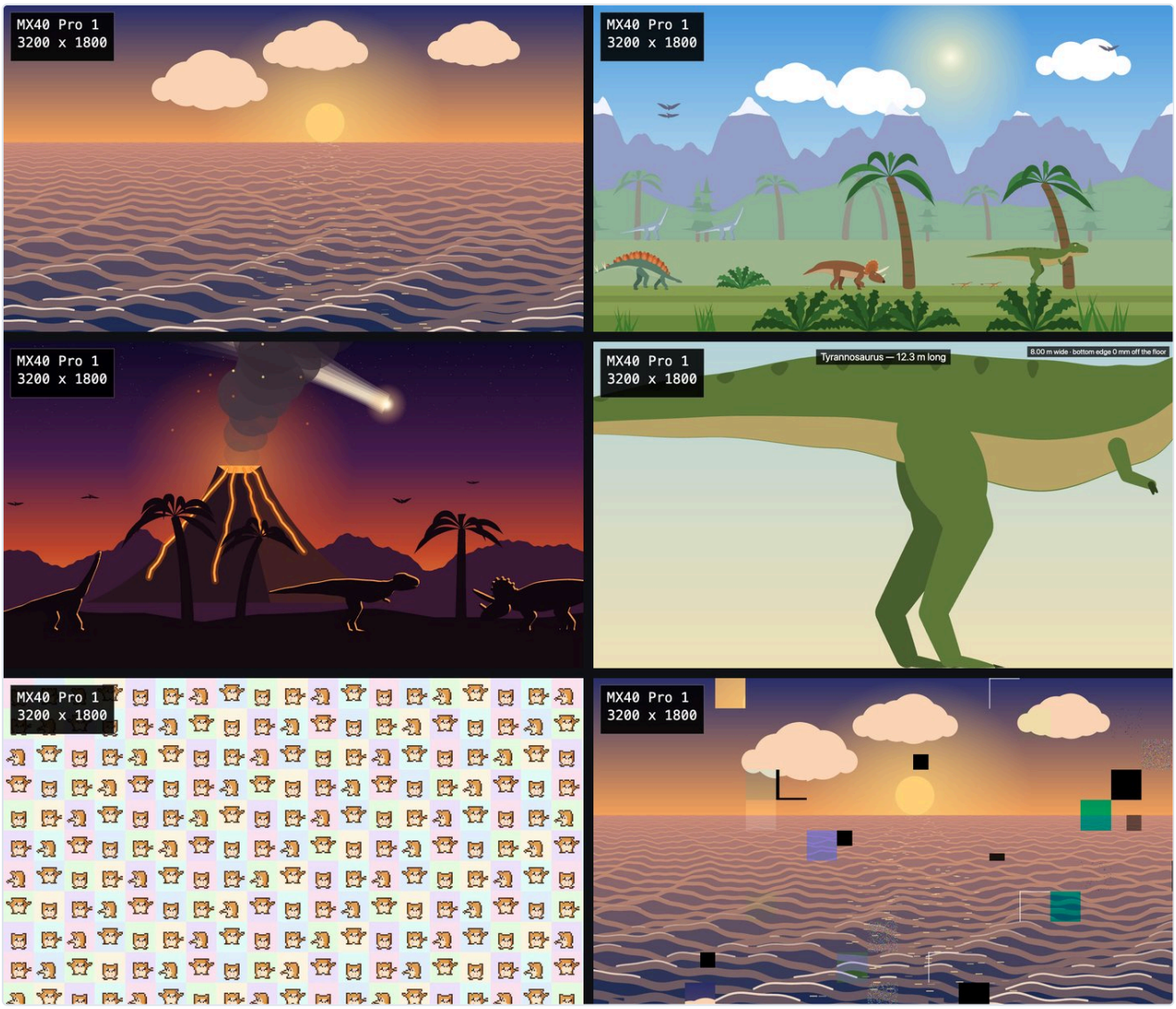
To measure something: play the MP4 through it and film the screen with sound (a phone will do), or record its output on anything that takes picture and sound together. Find where a beep starts in the sound, step to that frame, and read the lit box. **+2** means the sound arrives two frames after the picture; **−1**, one frame before it.

- The sound is in the **MP4**, and in **Live output**'s first window once you've clicked it. The preview is silent and a PNG is just a picture. Measure with the MP4: the live sound's delay through the browser hasn't been measured.
- Make the MP4 at the frame rate the system runs at, so a frame in the file is a frame on the screen.
- The middle of the ring takes the same **In the middle** settings as the classic card: the logo, your image, or your text.
- The file's first and last half second are quiet, so the loop's seam never falls in a beep.

How close it is: decoded by ffmpeg and by AVFoundation (what Resolume uses on a Mac), each beep starts within a quarter of a millisecond of the start of its flash frame, at 25, 29.97 and 30 fps. A frame is 33 to 40 ms, so the file itself is never the thing that's out.

Scale and novelty

The other sort of test pattern: the one that's up while the client walks in, or while you're explaining to someone what a dead module looks like.


MX40 Pro 1
3200 x 1800
MX40 Pro 1
3200 x 1800
MX40 Pro 1
3200 x 1800
MX40 Pro 1
3200 x 1800
Tyrannosaurus — 12.3 m long
8.00 m wide · bottom edge 0 mm off the floor
MX40 Pro 1
3200 x 1800
MX40 Pro 1
3200 x 1800

Pattern	What it is for
Rulers	Tape measures along the edges, metric or imperial, from the pixel pitch. Each tick is within half a pitch of true, and the pattern says how much that is. The side rulers can read height off the floor.
Banana for scale	A life-size bunch of five 19 cm bananas, once or on every LED tile, with the wall's size in bananas.
Life-size dinosaurs	Seven dinosaurs at their usual quoted sizes, a 1.75 m person and a banana, standing on the floor in a line. **Pan** along it, or pick one under **Centre on**. Anything bigger than the wall is cut off by its edges.
Hamster dance	One dancing pixel-art hamster per LED tile, at a tempo you set. Moves.
Ocean waves	A generative sea rolling in, day, sunset or night. **Another** picks a different sea. Moves.
Dinosaur scroll	A side-scrolling Cretaceous with dinosaurs walking on the ground that scrolls under them. Moves.
Volcano and comet	An eruption, an incoming comet, and some worried silhouettes. Moves.
Broken wall	Another pattern with a share of its tiles broken in fifteen different ways, from a dead tile to a wrong RGB order. Tick **Label each fault** to name them on the wall.
Your image	Any picture, fitted, filled and cropped, stretched, 1:1 or tiled. It stays in this browser across a refresh and is never uploaded.

The per-tile patterns (bananas per tile, hamsters, pumpkins, the broken wall) use the same tiles as the **LED tiles** pattern: a uniform size you state, or the cabinets from a Pixel Peeker import.

Halloween

Three more for the time of year when someone asks whether the wall can do "something spooky" between the bands.

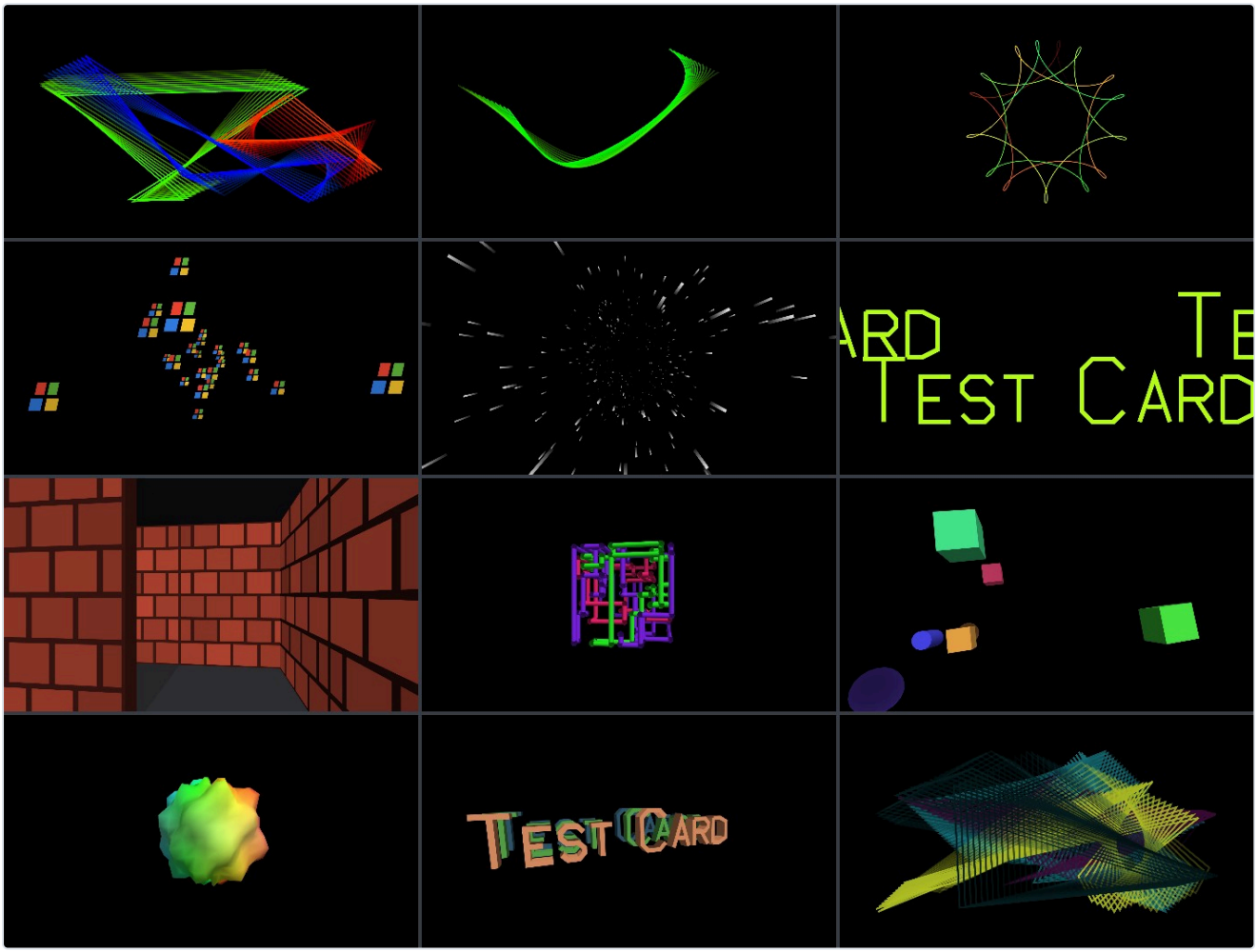


Pattern	What it is for
Pumpkin patch	A jack-o'-lantern on every LED tile, each carved with one of six faces: classic, fierce, happy, surprised, sly (with fangs) and goofy. Tick the faces you want; **Seed** deals them out. **Candles flicker** and **Blinks, winks and chomps** are the two things that move; untick both and it's a still. Untick every face and the pumpkins are plain.
Spooky forest	A night wood scrolling past in five layers of parallax: far hills, mist, three depths of dead trees (the middle one has eyes in it), lanterns on the ground, and witches and bats crossing the moon. **Seed** and **Speed** are shared with the dinosaur scroll. Moves.
Haunted house	A house on a hill with candles in its windows, a ghost in one of the dark ones once a loop, bats round the spire, vampires flying round the back, and zombies and skeletons walking the graveyard in opposite directions. **Seed** rearranges it. Moves.

Like the other scenes, the forest and the house carry on across outputs when **Carry on across outputs** is ticked. The zombies and skeletons go round once per loop, so a longer loop makes them walk slower; at 10 s they're a bit brisk for the undead.

Idle

The Windows 95 and 98 screensavers, for the hours a wall spends waiting for something to happen. They come from Idler, my Resolume generator of the same eleven, and they're drawn by the same code: a port of its savers, through its own two shaders.



Pattern	What it is for
Mystify	Polygons bouncing round the screen, each trailing its own recent shapes. Moves.
Beziers	Mystify's motion with a curve through the points. Moves.
Curves and Colors	A spirograph drawing itself on a rotating palette. Moves.
Flying Windows	Four-pane logos streaming out of the middle of the screen. Moves.
Flying Through Space	The starfield. Turn up **Streak** for warp speed. Moves.
Scrolling Marquee	Your **Text**, crossing the screen and bobbing as it goes. Moves.
3D Maze	A first-person walk down brick corridors, in a maze with no end to it. Moves.
3D Pipes	Plumbing growing into a box, the odd teapot included. Moves.
3D Flying Objects	Lit solids, boxes to teapots, tumbling past. Moves.
3D FlowerBox	A solid on the spot, swelling between a box, a ball and a flower. Moves.
3D Text	Your **Text** in extruded letters, turning. Moves.

Each one starts as it ran on the machine and keeps its own settings, so fiddling with Mystify leaves the maze alone; **As it ran** puts one back. The sliders are Idler's. Its six scene controls mean something different to each saver, so here they're named for what they do in that one: **Polygons**, **Corners** and **Trail** in Mystify, **Pipes**, **Grid** and **Fill before clearing** in 3D Pipes. The 3D ones also have **Shading** (lit or wireframe), a camera and **Fog**. Every one has **Speed**, which stops it altogether at the bottom, and a **Colour** mode: the saver's own palette, one colour, a spread of hues, or one colour rotating.

A loop has to come back to where it started, and the originals never do. So every speed in them is rounded to a whole number of cycles in the loop: on a 10 s loop the Mystify corners each make one or two trips and 3D FlowerBox turns over once. A longer loop leaves more of the original speeds alone. 3D Pipes and 3D Maze grow, and nothing rounds that, so each loop is one run from the **Seed**, starting again once a loop: the box clears, as it used to when it filled up, and the maze fades out and back in at a new start. The restart falls part-way through the loop, so the still at 0 s is well into a run, not an empty box.

With **Carry on across outputs** ticked they're one picture across the whole surface, each output a window onto it: the maze is seen wider across a span, not closer.

They need WebGL2, which any recent browser has; without it the pattern says so instead of drawing.

Physical scale: give it a pitch

The rulers, the bananas and the dinosaurs are drawn at real size, so they need to know how big a pixel is.

- A **Pixel Peeker** import brings the wall's pitch with it.
- For anything else, open **Edit** on the output and type the **pixel pitch** in millimetres, or the image's **width in metres** and let it work the pitch out. For a projector, the width is the one you'll know.
- An output with neither is drawn at the **fallback pitch** in the pattern panel and says **pitch assumed** across the bottom. The ZIP manifest lists each output's pitch and marks the guesses.

Bottom edge above the floor puts the dinosaurs' feet on the real floor rather than on the bottom of the wall. A wall flown a metre up shows their knees, not their toes.

Carry on across outputs treats the outputs as windows onto one surface, placed by where they sit in the composition, so a ruler runs the whole wall and the sea has no seam between two projectors. It only applies to the scale and novelty patterns; bars and grids always belong to one output, and LED tiles stay with the output whose cabinets they are. An output stretched to a different shape in the composition isn't spanned and gets its own copy instead.

Moving patterns: MP4 loops

The moving patterns animate in the preview (**Pause** freezes it on the frame the PNG will use) and get a **Download MP4 loop** button. Set the length and frame rate under **Loop**.

Download PNG is still there for every one of them: it gives you a single still per output, the frame at **PNG shows the frame at (s)**. Pause the preview to see which frame that is.

- The loop goes round with no seam: everything in it does a whole number of cycles in the loop's exact length, which is a whole number of frames.
- It's H.264, encoded in your browser. That needs WebCodecs; I've only tried it in Chrome.
- Colour is BT.709, limited range, converted here and labelled in the file. Legal-range bars keep their sub-black and super-white.
- **Bitrate** at 0 picks a generous one for the raster. Set your own if the files are too big.
- Big rasters take a while: the export shows its progress and has a **Cancel**. It keeps going at the same speed behind another tab.
- A loop that would come to more than about 1 GB is refused before it starts, since the whole file has to fit in the browser tab's memory.
- **Too big for one video:** Chrome stops at 4096 × 2304 worth of picture, in any shape, so a bigger raster comes out as a grid of **tiles**: 8K is four 3840 × 2160. A note above the preview says which outputs will be tiled and how, and the button says it's a ZIP. Each tile is its own

seamless loop of the same length; the manifest and each file name (`…_tile-r1c2_3840x2160+3840+0.mp4` : row, column, size, then x and y) say where it goes. Place them at those positions and play them together. A bitrate you've set is shared between the tiles; auto is worked out per tile.

Endless, for live output. The generative patterns (the idle savers, the hamsters, the ocean, the dinosaur scroll, the volcano, the pumpkin patch, the spooky forest and the haunted house) have one more switch under **Loop**: **Endless, for live output**. A loop has to come back round to where it started; a live output doesn't, so with this on nothing is fitted to the loop and nothing comes round twice:

- Every bat, witch, vampire and pterosaur is a new one each time it crosses: where it flies, its size, how it looks. In the graveyard each one walking past is a zombie or a skeleton afresh, at its own height and nearness. Nothing changes while it's on the picture.
- Each smoke puff and lava bomb is new, each band of the sea rolls in with a shape of its own, each hamster tile picks a new dance every time round, the ghost picks a different window, and the twinkles, glows and sways drift out of step with each other.
- The idle savers run at Idler's own speeds, and the stars and windows flying out of the middle come out on new paths each time. 3D Pipes and 3D Maze build a new network from a seed of their own every couple of minutes of the saver's time, so they never build the same one twice.
- The scenery (the hills, the trees, the stones) still scrolls past on its own period, and the bats circling the haunted house's spire are the same seven, never in the same places twice. It's what moves through the scenery that's new.

Live output and the preview run it for ever, and every window works out where it is from the computer's clock, so a span stays in step. Because it isn't a loop any more, the button becomes **Download MP4 slice**: the loop's length of the run starting at **PNG shows the frame at (s)**, named `…_slice.mp4` . It's for a preview or a social post, not a media server: played round, it jumps. AV sync doesn't get the switch. It's a measurement, and it has to be a loop.

In Resolume. Arena 7.28.1 plays them at the right size, rate and length, with the right colour, and loops them without dropping or repeating a frame. The catch: an H.264 clip triggered from cold often plays for about half a second and then holds that picture until its first wrap. That's Arena's H.264 playback, not this file (ffmpeg's own H.264 encodes do the same; DXV doesn't). Before a show, convert the loop to **DXV** with Alley, which comes with Arena, or trigger the clip and let it run once round.

Two ways to render, and when each is the right one

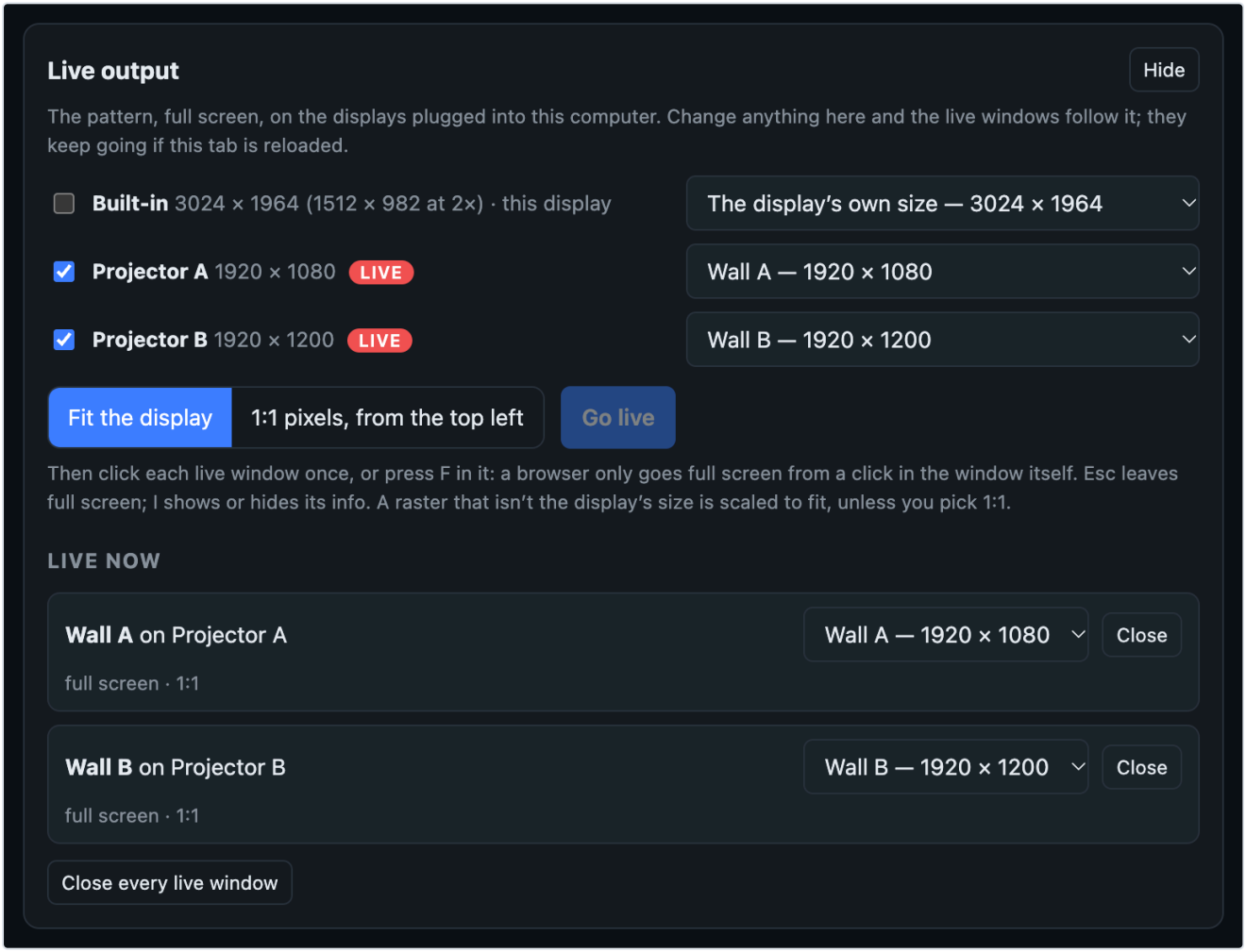
One PNG per output, each at that output's native raster. One output downloads as a bare PNG; several download as a ZIP with a `manifest.txt` recording which file is which output, the level range used, and any raster the app had to infer.

Single composition canvas: one image at composition size with every output's region drawn and labelled. Play it through Resolume and each physical output should light up showing its own name.

That second one is the fastest proof the advanced output map matches what you think it does, and you don't need any way of getting a file to each output individually, which on a show day is often the actual problem.

Live output: straight to the displays

Live output, beside the download buttons, puts the pattern full screen on the displays plugged into this computer, with no file in between. It's for when the laptop is the source: plugged into the projectors or the LED processor, or bars on a monitor before anything else is rigged.



Three displays listed. Each projector got the output that's its size, and both are live, full screen at 1:1; the laptop's own screen is left out.

1. Press **Live output**. In Chrome or Edge, **Find my displays** asks whether the page may manage windows on all your displays. Say yes and it lists them, each with its raster in device pixels. On Windows at 125% or 150% the browser only knows that to a pixel or so, and says "about".

2. Tick the displays to use and choose what each one **Shows**: an output from your list, the whole composition, or **The display's own size**, which needs no import at all. An output that's the same size as a display goes to that display by itself.
3. Press **Go live**. A window opens over each display.
4. Click each window once, or press **F** in it, and it goes full screen on its display.

Browsers won't let a page do two things, which is why there's clicking involved:

- **One window per click.** The first display's window opens and the panel says which didn't. Press **Go live** again for each one left, or allow pop-ups for the site (the icon at the right of the address bar) and they'll all open at once.
- **Full screen needs a click in the window itself.** Nothing the editor does can do it for you.

Once they're up:

- The live windows follow the editor. Change the pattern, the burn-in, the outputs or the picture and they change with it. Each is a page of its own, so reloading or closing the editor tab leaves them running, and the editor finds them again when it comes back.
- **Live now** lists each window: what it shows, whether it's full screen, 1:1 or scaled, and the frame rate of a moving pattern. You can change what one shows, mute it, or close it from there.
- **Fit the display** scales a raster that isn't the display's size until it fills it, and the window says by how much (scaled, 1px detail is approximate). **1:1 pixels, from the top left** never scales, and crops a raster that's bigger than the display.
- In a window, **Esc** leaves full screen and **I** shows or hides the info. The pointer hides itself once it's full screen.
- Moving patterns play live. Every window works out where it is in the loop from the computer's clock, so a scene spanning several outputs stays in step across them. An endless one (see **Moving patterns**) has no loop to be in: each window works out where it is in the run from the same clock, so they still agree, and never come round.
- AV sync's beep comes from the first window opened, once it's been clicked (a browser won't make a sound before that). The sound is put back in step with the picture every time round the loop.
- A live window stops the display going to sleep. Where the browser won't allow that (on plain http, for one), the window says so, and you'll want display sleep off by hand.

Without the display list (Safari, Firefox, the permission refused, or a copy opened over plain http from another machine, as the desktop app's is), **Open a live window** opens one here. Drag it to its display and click it.

A Mac at a scaled resolution. Set a display to anything but its default in System Settings and macOS draws a bigger picture than the display takes, then shrinks it to fit the signal. The browser only sees the bigger picture, so the panel lists that size, and "1:1" in the window means 1:1 with it,

not with the cable. Put the display on its default resolution for a pixel check. Projectors and LED processors usually come up that way anyway.

What it can't tell you. I haven't measured what comes out of the computer's video port. The browser and the operating system colour-manage everything on screen to the display's colour profile, so the code values on the cable may not be the PNG's. Live output is for checking the map, the geometry, the scaling and the pixel check; for levels, the pluge and exact bar values, play the PNG or MP4 from the media server. The same goes for AV sync: the browser adds its own delay to the picture and to the sound, which nobody has measured, so to measure a system, play the MP4.

Level range: the app asks because a PNG cannot say

SMPTE bars are defined in Y′CbCr. A PNG is RGB, and a PNG can't record which convention it was written in. So the app asks, and writes the answer into the manifest.

- **Full 0–255** (default): media server, GPU output, HDMI/DP set to full range. **The −2% pluge patch clips to black and can't be judged.**
- **Legal 16–235**: studio swing, for an SDI chain. Sub-black and super-white survive, so the pluge means something. It looks washed out on a full-range display, and it's meant to.

Pick the one that matches the chain the file is going down, not the one that looks right on your laptop.

Imports, and what each file does not know

Resolume: one output per screen. The raster is read from the output device when it records one, and otherwise **inferred from slice bounds, which can come out too small.** Inferred rasters are badged in the UI and called out in the manifest, so correct them by hand.

Pixel Peeker: use **Export → JSON**, not *Save project*. The project file stores cabinets as library references in millimetres and can't be resolved to pixels here, so it's refused with a message naming the right export, rather than half-imported.

Blend Calc: needs the projector resolution (not in the file) and, for designs using a fit mode, the **solved** overlap (also not in the file, because Blend Calc works it out and never writes it back). Both are asked for. Its Resolume XML export needs neither and is the better file to bring.

You can also just add outputs by hand.

If something is wrong

Symptom	Cause
The pixel-check pattern is flat grey on the output	Something in the chain is scaling. That is exactly what the pattern is for.
An output's raster looks too small	It was inferred from slice bounds. It is badged; correct it by hand.
A Pixel Peeker file is refused	It is a project save, not the JSON export.
The pluge patch is invisible	Full-range levels clip it to black. Render at legal range for an SDI chain.
The bars look washed out	Legal-range content on a full-range display. Correct, not broken.
Blend Calc import asks for numbers	The projector resolution and the solved overlap are not in that file. Bring its Resolume XML instead.
"Pitch assumed" across the bottom	The output has no pixel pitch. Set it under **Edit**, or the life-size things are only as right as the fallback.
The dinosaurs' feet are missing	The wall's bottom edge is off the floor, and so are their feet. Set **Bottom edge above the floor** to 0 to see them.
No MP4 button	The pattern doesn't move. A broken wall moves if its picture does, or with flicker or data noise ticked.
"H.264 needs an even width and height"	4:2:0 video can't carry an odd raster. Export PNGs, or check the raster.
The MP4 comes as a ZIP of tiles	The raster is bigger than this browser will encode as one video. Place each tile where its name and the manifest say.
"...will not take ... Mb/s at ..." or **"... Mb/s is more than H.264 allows"**	The bitrate is too high for that raster. Lower it, or set 0 for auto.
"...will not make H.264 at ... fps at any size"	This browser's encoder can't do that frame rate. Try another, or a current Chrome or Edge, or export PNGs.
"...will not take ..., whole or cut into as many as 64 tiles"	Too big for this browser's encoder even in 64 pieces. Export PNGs.
An MP4 freezes after half a second in Resolume, then carries on at the wrap	Arena's H.264 playback from cold, not the file. Convert it to DXV with Alley, or let it run once round before the show.
"That loop would be about ... GB" or **"Those ... loops would come to about ... GB"**	Too long, too big or too high a bitrate to build in a tab, every tile and output in the ZIP counted. Shorten the loop, lower the bitrate, or export fewer outputs at a time.
Only one live window opened	A browser lets a page open one window per click. Press **Go live** again, or allow pop-ups for the site.

Symptom	Cause
A live window won't go full screen	It needs a click, or F, in that window. The editor can't do it for you.
"scaled ...%" in a live window	The raster isn't the display's size. Give it the output that matches, **The display's own size**, or pick 1:1.
Live output has no display list, only "Open a live window"	Not Chrome or Edge, the permission was refused, or the page is on plain http. Drag the window to its display instead.
"can't stop the display sleeping here"	The browser wouldn't hold a wake lock (it needs https or localhost). Turn display sleep off by hand.
A live 1:1 pixel check is soft on a Mac	The display is on a scaled resolution, so macOS is resizing it. Put it back on its default.
Live bars measure wrong on a scope	The browser's and the OS's colour management. For levels, play the PNG from a media server.
The image has gone	It's kept in this browser's storage. Another browser, a private window or cleared site data won't have it.